

Portfolio Optimization & Factor Analysis

Mean-variance optimization, benchmark comparison, and factor-model analysis

Objective

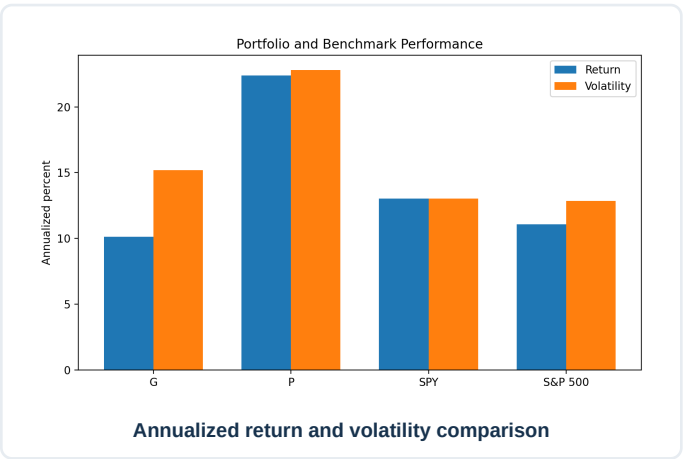
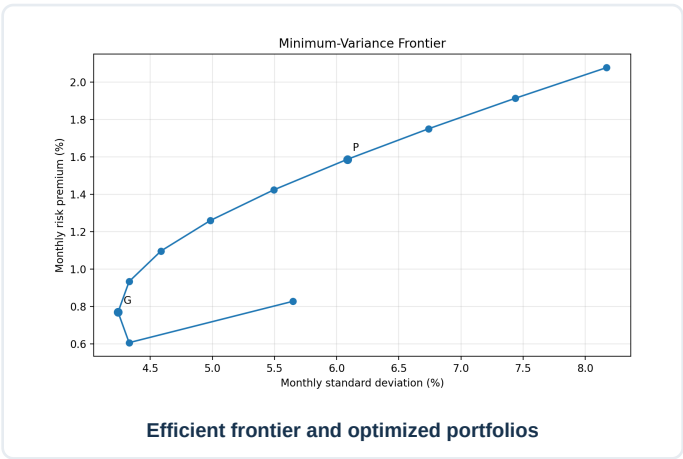
Construct global minimum-variance and maximum-Sharpe portfolios from eight U.S. equities, map a ten-point efficient frontier, compare performance with SPY and the S&P 500, and evaluate the optimized portfolio using common asset-pricing models.

Data and methodology

Monthly adjusted closing prices from January 2015 through December 2023 were converted to holding-period returns. Excel Solver was used to estimate a covariance-based global minimum-variance portfolio, a maximum-Sharpe portfolio, and eight additional frontier portfolios. Stata was used to estimate a market model, the Fama-French three-factor model, and the Carhart four-factor model.

Headline findings

- Maximum-Sharpe portfolio: 22.4% annualized return, 22.8% volatility, and 0.925 annualized Sharpe ratio.
- SPY: 13.0% annualized return, 13.0% volatility, and 0.900 annualized Sharpe ratio.
- The optimized portfolio's modest Sharpe advantage required short positions and concentrated weights.
- Fama-French and Carhart models explained approximately 2%-3% of monthly optimized-portfolio return variation.
- The market model explained approximately 28% of monthly optimized-portfolio excess-return variation.



Interpretation

The maximum-Sharpe portfolio delivered the strongest in-sample risk-adjusted performance, but its advantage over SPY was small relative to its greater volatility and implementation complexity. The low multifactor R-squared values indicate that size, value, and momentum exposures explained little of the optimized portfolio's observed monthly return variation.